

EXPLORATORY DATA ANALYSIS LAB

Experiment 1 - 7/07/2026

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1. Write a python program to print the factorial of a number.

CODE

```
n = int(input("enter a number you want to calculate the factorial of :"))
if n<0:
    print("enter a positive number.")
elif n!=0:
    fact = 1
    for i in range(1,n+1):
        fact = i*fact
    print("factorial of", n, "is", fact)
else:
    print("factorial of 0 is 1.")
```

OUTPUT

```
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on li
Type "help", "copyright", "credits" or "license()" for more in
>>> print("Riya Stanley 25BAI0258")
Riya Stanley 25BAI0258
>>>
===== RESTART: /home/anaconda/fact.py =====
enter a number you want to calculate the factorial of :5
factorial of 5 is 120
>>>
===== RESTART: /home/anaconda/fact.py =====
enter a number you want to calculate the factorial of :0
factorial of 0 is 1.
>>>
===== RESTART: /home/anaconda/fact.py =====
enter a number you want to calculate the factorial of :-6
enter a positive number.
>>>
```

2. Write a python program to rotate a given array by one element

CODE

```
arr=list(map(int,input("enter a list of numbers:").split()))

n = len(arr)

l=arr[0]

for i in range(n-1):

    arr[i]=arr[i+1]

arr[n-1]=l

print("rotated array is", arr)
```

OUTPUT

```
>>> ===== RESTART: /home/anaconda/rotate.py
enter a list of numbers: 2 3 4 5 6
rotated array is [3, 4, 5, 6, 2]
>>> print("Riya Stanley 25BAI0258")
Riya Stanley 25BAI0258
>>>
```

3. Write a python program to calculate the sum of even and odd position elements

CODE

```
n=list(map(int,input("enter a list of numbers:").split()))
```

```
esum=0
```

```
osum=0
```

```
for i in range(len(n)):
```

```
if (i+1)%2==0:
```

```
    esum+=n[i]
```

```
else:
```

```
    osum+=n[i]
```

```
print("sum of even digits is :",esum)
```

```
print("sum of odd digits is :",osum)
```

OUTPUT

```

>>> ===== RESTART: /ho
enter a list of numbers:2 3 4 5 6
sum of even digits is : 8
sum of odd digits is : 12
>>> print("Riya Stanley 25BAI0258)
...
SyntaxError: unterminated string lite
>>> print("Riya Stanley 25BAI0258")
...
Riya Stanley 25BAI0258
>>>

```

4. Write a python program to display the grade for a given score.

CODE

```
marks = int(input("enter marks: "))
```

```
if marks>=90 and marks<=100:
```

```
    print("Grade A")
```

```
elif marks>=80 and marks<=89:
```

```
    print("Grade B")
```

```
elif marks>=70 and marks<=79:
```

```
    print("Grade C")
```

```
elif marks>=60 and marks<=69:
```

```
    print("Grade D")
```

```
elif marks>=50 and marks<=59:
```

```

    print("Grade E")
elif marks>=40 and marks<=49:
    print("Grade F")
else:
    print("FAIL.")

```

OUTPUT

```

===== RESTART: C:/Users/Riya/grades.py =====
enter marks: 70
Grade C
>>>
===== RESTART: C:/Users/Riya/grades.py =====
enter marks: 88
Grade B
>>>
===== RESTART: C:/Users/Riya/grades.py =====
enter marks: 92
Grade A
>>>
===== RESTART: C:/Users/Riya/grades.py =====
enter marks: 33
FAIL.
>>> print("Riya Stanley 25BAI0258")
Riya Stanley 25BAI0258
...

```

5. Write a python program to count the number of times a word appears in a string using dictionary

CODE

```

text=input("Enter a list of words:")
w = text.split()
count = {}
for i in w:
    if i in count:
        count[i] += 1

```

else:

count[i] = 1

print(count)

OUTPUT

```
===== RESTART: C:/Users/Riya/dict.py =====  
Enter a list of words:apple banana apple cherry banana apple  
{'apple': 3, 'banana': 2, 'cherry': 1}  
>>> print("Riya Stanley 25BAI0258")  
Riya Stanley 25BAI0258  
>>>
```

6. Write a python program to check if a string is a palidrome.

CODE

```
def is_palindrome(text):
```

```
    l = 0
```

```
    r = len(text) - 1
```

```
    while l < r:
```

```
        if text[l] != text[r]:
```

```
            return False
```

```
        l += 1
```

```
        r -= 1
```

```
    return True
```

```
word = input("Enter a word: ")
```

```
if is_palindrome(word):
```

```
    print(word,"is a palindrome.")
```

```
else:
```

```
    print(word,"is not a palindrome.")
```

OUTPUT

```

=====
Enter a word: malayalam
malayalam is a palindrome.
>>> print("Riya Stanley 25BAI0258")
Riya Stanley 25BAI0258
>>>

```

7. Write a python program to add the matrices.

CODE

```

a = [[3, 9, 5], [0, 8, 3], [5, 0, 9]]
b = [[9, 0, 9], [5, 7, 9], [1, 2, 3]]

```

```

res = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]

```

```

for i in range(len(a)):
    for j in range(len(a[0])):
        res[i][j] = a[i][j] + b[i][j]

```

```

for r in res:
    print(r)

```

OUTPUT

```

=====
[12, 9, 14]
[5, 15, 12]
[6, 2, 12]
print("Riya Stanley 25BAI0258")
Riya Stanley 25BAI0258

```

8. Write a python program to make a simple calculator.

CODE

```

a=float(input("enter the first number:"))
b=float(input("enter the second number:"))

```

```

while(True):
    print("enter 1. for addition 2. for subtraction 3. for multiplication 4. division 5. exit")

```

```

choice = int(input("enter choice:"))
if choice == 1:
    print("sum is ", a+b)
elif choice == 2:
    print("subtraction is ", a-b)
elif choice == 3:
    print("multiplication is ", a*b)
elif choice == 4:
    print("division is ", a/b)
elif choice == 5:
    print("Exit.")
    break
else:
    print("Invalid choice")

```

OUTPUT

```

===== RESTART: C:/Users/Riya/calcul.py =====
enter the first number:6
enter the second number:4
enter 1. for addition 2. for subtraction 3. for multiplication 4. division 5. exit
enter choice:1
sum is 10.0
enter 1. for addition 2. for subtraction 3. for multiplication 4. division 5. exit
enter choice:2
subtraction is 2.0
enter 1. for addition 2. for subtraction 3. for multiplication 4. division 5. exit
enter choice:3
multiplication is 24.0
enter 1. for addition 2. for subtraction 3. for multiplication 4. division 5. exit
enter choice:4
division is 1.5
enter 1. for addition 2. for subtraction 3. for multiplication 4. division 5. exit
enter choice:5
Exit.
>>>

```

9. Write a python code to find the number of vowels in a string

CODE

```

t=input("enter a word: ")
count=0
for i in t:
    if i in ["a", "e", "i", "o", "u", "A", "E", "I", "O", "U"]:

```



```
count+=1
```

```
print("the number of vowels in the word are : ", count)
```

OUTPUT

```
>>>===== RESTART: C:/Users/Riya/vowels.py =====
enter a word: multiplication
the number of vowels in the word are : 6
>>>===== RESTART: C:/Users/Riya/vowels.py =====
enter a word: DIVISION
the number of vowels in the word are : 4
>>>|
```

10. Write a python program to remove all repetitions of all elements in a list.

CODE

```
m = list(map(int,input("enter a list of numbers (repeation allowed) : ").split()))
new = []
for i in m:
    if i not in new:
        new.append(i)
print("New list without repeated elements : ", new)
```

OUTPUT

```
===== RESTART: C:/Users/Riya/repeatations.py =====
enter a list of numbers (repeation allowed) : 2 4 3 2 3 4 6 7
New list without repeated elements : [2, 4, 3, 6, 7]
>|
```